

Group II, Row IV — the p^+ integration

Can the plus prescription be used, and the calculation carried to the end

$$S = p^+ + k^+, \quad t = \frac{p^+}{k^+}, \quad r = y' - y, \quad \kappa = k_\perp \cdot r, \quad X = \frac{x - y}{(x - y)^2}, \quad X' = \frac{x' - y'}{(x' - y')^2}, \quad \mathcal{L} = \frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu_{\text{MS}}^2 r^2)$$

Answer: yes, and cleanly. The only $p^+ \rightarrow 0$ singularity in this row is the $\frac{1}{p^+} \delta_{mm'} \delta_{kk'}$ term, and its companion function is the phase $F(p^+) = e^{-i\kappa p^+/k^+}$, which is perfectly regular at $p^+ = 0$ with $F(0) = 1$. That is exactly the condition the plus prescription needs, so

$$\int_{\Lambda}^{\vee-k^+} \frac{dp^+}{p^+} F(p^+) = \underbrace{F(0) \log \frac{\vee - k^+}{\Lambda}}_{\text{rapidity log} \rightarrow \text{JIMWLK}} + \underbrace{\int_0^{\vee-k^+} dp^+ \frac{F(p^+) - F(0)}{p^+}}_{\text{finite, stays in the impact factor}}$$

works with no extra subtraction — unlike the running-coupling row, where the companion carried its own $\log \bar{\xi}$ and the naive $[\]_+$ failed. **One warning, which is new:** the $\frac{1}{\epsilon}$ in $\mathcal{L}$ is **infrared** (from $|z| \rightarrow \infty$), not ultraviolet — see §5.

1. The contraction, once more

$\mathcal{G}^{mm'}$ is symmetric, so the $\frac{2}{k^+} (\delta_{mk'} \delta_{m'k} - \delta_{m'k'} \delta_{km})$ term contracts to zero. Keeping $d_\perp = 2 - 2\epsilon$ and using $\epsilon \mathcal{L} \rightarrow 1$,

$$\begin{aligned} \delta^{mm'} \mathcal{G}^{mm'} &= \pi(\mathcal{L} - 2), \quad d_\perp X_m X'_m \mathcal{G}^{mm'} = \pi(\mathcal{L} - 1)(X \cdot X') - \frac{2\pi(X \cdot r)(X' \cdot r)}{r^2}, \\ \implies [\text{bracket}]^{kmk'm'} \mathcal{G}^{mm'} X^k X'^{k'} &= \pi \left[\frac{p^+}{S^2} A + \left(\frac{p^+}{k^{+2}} + \frac{1}{p^+} \right) B \right], \end{aligned}$$

$$\boxed{A = (\mathcal{L} - 1)(X \cdot X') - \frac{2(X \cdot r)(X' \cdot r)}{r^2}, \quad B = (\mathcal{L} - 2)(X \cdot X')}$$

Neither A nor B depends on p^+ . That is the whole reason the p^+ integral factorises into two numbers.

2. The two p^+ masters

Use $t = \frac{p^+}{k^+}$, so $\sigma \equiv \frac{S}{k^+} = 1 + t$, $dp^+ = k^+ dt$, and the phase of your expression is $e^{-i\kappa} e^{-i\kappa t}$. Then $\frac{p^+}{S^2} = \frac{1}{k^+} \frac{t}{(1+t)^2}$ and $\frac{p^+}{k^{+2}} + \frac{1}{p^+} = \frac{1}{k^+} \left(t + \frac{1}{t} \right)$, so with $\frac{k^+}{2\pi} \cdot \pi \cdot \frac{1}{k^+} = \frac{1}{2}$:

$$\begin{aligned} \int_{\Lambda}^{\vee-k^+} \frac{dp^+}{2\pi} e^{-i\sigma\kappa} [\dots] &= \frac{1}{2} [A J_1 + B J_2], \\ J_1 &= e^{-i\kappa} \int_0^\infty dt \frac{t e^{-i\kappa t}}{(1+t)^2}, \quad J_2 = e^{-i\kappa} \int_\lambda^T dt \left(t + \frac{1}{t} \right) e^{-i\kappa t}, \end{aligned}$$

with $\lambda = \frac{\Lambda}{k^+}$ and $T = \frac{\vee}{k^+} - 1$.

J_1 : convergent at both ends, no prescription needed

In the variable $\sigma = 1 + t$ it is $\int_1^\infty \left(\frac{1}{\sigma} - \frac{1}{\sigma^2} \right) e^{-i\kappa\sigma} d\sigma = E_1(i\kappa) - E_2(i\kappa)$, and $E_2(z) = e^{-z} - z E_1(z)$ gives

$$\boxed{J_1(\kappa) = (1 + i\kappa) E_1(i\kappa) - e^{-i\kappa}}$$

(checked against oscillatory quadrature to 10 digits at $\kappa = 0.4, 1.2, 3.0$). Small κ : $J_1 \rightarrow -\gamma_E - \log(i\kappa) - 1$.

3. J_2 and the plus prescription

The criterion

$\int_\lambda \frac{dt}{t} g(t)$ may be written as $g(0) \log \frac{1}{\lambda} + \int_0 [\frac{1}{t}]_+ g$ **if and only if** $g(0)$ **exists**. Here $g(t) = e^{-i\kappa t}$ and $g(0) = 1$: the p^+ dependence sits only in the phase, and the phase is analytic at $p^+ = 0$. Nothing else in the integrand is singular there ($\frac{p^+}{S^2} \rightarrow 0$, $\frac{p^+}{k^{+2}} \rightarrow 0$, and $A, B, \mathcal{L}$ are p^+ -independent). **So the prescription applies.**

$$\frac{\theta(t-\lambda)}{t} = \left[\frac{1}{t} \right]_+ + \delta(t) \log \frac{1}{\lambda} + O(\lambda), \quad \int_0^T dt \left[\frac{1}{t} \right]_+ f(t) \equiv \int_0^T dt \frac{f(t) - f(0)}{t}$$

Numerical confirmation of the split (direct integral vs plus-prescription form):

κ	λ	T	direct	plus form
0.7	10^{-5}	30	$11.333275 - 1.594884 i$	$11.333275 - 1.594891 i$
2.0	10^{-6}	15	$12.512115 - 1.566755 i$	$12.512115 - 1.566757 i$
0.2	10^{-7}	80	$17.136118 - 1.631302 i$	$17.136118 - 1.631302 i$

Doing the two pieces

Subtracted (the $[\]_+$ piece), in closed form:

$$\int_0^T dt \frac{dt}{t} [e^{-i\kappa t} - 1] = -\gamma_E - \log(i\kappa T) - E_1(i\kappa T)$$

(verified to 10 digits). Linear piece, exact and regular at $\kappa \rightarrow 0$:

$$\int_0^T t e^{-i\kappa t} dt = \frac{e^{-i\kappa T}(1 + i\kappa T) - 1}{\kappa^2} \xrightarrow{\kappa \rightarrow 0} \frac{T^2}{2}$$

Assembling, and noting that $\log \frac{T}{\lambda} - \log(i\kappa T) = \log \frac{1}{\lambda} - \log(i\kappa)$ so that the longitudinal cutoff **drops out of the logarithm**:

$$J_2(\kappa) = e^{-i\kappa} \left[\log \frac{k^+}{\Lambda} - \gamma_E - \log(i\kappa) - \frac{1}{\kappa^2} \right] + e^{-i\kappa} \left[\frac{e^{-i\kappa T}(1 + i\kappa T)}{\kappa^2} - E_1(i\kappa T) \right]$$

The second bracket oscillates in κ with period $\sim 1/T$ and is annihilated by the remaining $\int_{y,y'}$ (Riemann–Lebesgue) as $\Lambda \rightarrow \infty$. Keeping only the first,

$$J_2(\kappa) = e^{-i\kappa} \left[\log \frac{k^+}{\Lambda} - \gamma_E - \log |\kappa| - i \frac{\pi}{2} \text{sgn } \kappa - \frac{1}{\kappa^2} \right]$$

Why the $\frac{p^+}{k^+}$ term is not a plus-prescription problem but does need the phase. It vanishes at $p^+ = 0$, so it is invisible to $[\]_+$. At the *upper* end it grows linearly; drop the $e^{-i\kappa p^+/k^+}$ from your phase and it gives $\frac{\sqrt{2}}{2k^+}$, a quadratic divergence in the longitudinal cutoff. With the phase it is the bounded oscillating term above. Your phase — $e^{-ik(y'-y)} e^{-ik(y'-y)p^+/k^+}$ — is now correct, and this is what it buys you.

4. The finished cross section

$$\begin{aligned} \frac{d^3 N^{\text{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4}{4\pi^4} \frac{N_c}{2k^+} \int_{x,x',y,y'} \left\{ J_1(\kappa) A + J_2(\kappa) B \right\} [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x) \\ A &= (\mathcal{L} - 1)(X \cdot X') - \frac{2(X \cdot r)(X' \cdot r)}{r^2}, \quad B = (\mathcal{L} - 2)(X \cdot X'), \\ J_1 &= (1 + i\kappa) E_1(i\kappa) - e^{-i\kappa}, \quad J_2 = e^{-i\kappa} \left[\log \frac{k^+}{\Lambda} - \gamma_E - \log(i\kappa) - \frac{1}{\kappa^2} \right], \\ \mathcal{L} &= \frac{1}{\epsilon} + 2 - \log(4\pi^2 \mu_{\text{MS}}^2 r^2), \quad r = y' - y, \quad \kappa = k_\perp \cdot r. \end{aligned}$$

Written with the rapidity log separated out, which is the form you want for matching onto evolution:

$$J_1 A + J_2 B = \underbrace{e^{-i\kappa} (\mathcal{L} - 2)(X \cdot X') \log \frac{k^+}{\Lambda}}_{\text{rapidity divergent}} + \underbrace{J_1 A + e^{-i\kappa} \left[-\gamma_E - \log(i\kappa) - \frac{1}{\kappa^2} \right] B}_{\text{the } [\]_+ \text{ remainder: finite, } \Lambda\text{-independent}}$$

5. Read the $\frac{1}{\epsilon}$ correctly: it is infrared

This is the one thing that differs from the running-coupling row, and it changes what you do with the pole. The divergence of $\mathcal{G}^{mm'}$ is **not** at $z \rightarrow y$. Trace it:

$$\delta^{mm'} \mathcal{G}^{mm'} = \int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2}$$

- $z \rightarrow y$ (or $z \rightarrow y'$): integrand $\sim \frac{1}{|y-z|}$, integrable in two dimensions. **No UV.**
- $|z| \rightarrow \infty$: integrand $\rightarrow \frac{1}{z^2}$, and $\int^\infty \frac{d^2 z}{z^2}$ is log divergent. **This is the pole.**

With a hard cutoff $|z| < R$, the trace is $\pi \log \frac{R^2}{r^2}$ — verified numerically:

$R/ r $	20	60	180	540
numeric	18.8227	25.7301	32.6331	39.5391
$\pi \log(R^2/r^2)$	18.8227	25.7255	32.6283	39.5311

So in $d_\perp = 2 - 2\epsilon$ it is $\epsilon > 0$ that regulates it, and

$$\mathcal{L} = \frac{1}{\epsilon_{\text{IR}}} + 2 - \log(4\pi^2 \mu_{\text{MS}}^2 r^2).$$

Consequences, all of which matter for what you do next:

- **Do not absorb it into the running coupling.** $\overline{\text{MS}}$ here is only a bookkeeping convention for the constant that rides with $\log r^2$; there is no charge renormalisation to do.
- **It must cancel against companion rows.** $\int d^2 z \frac{(y-z) \cdot (y'-z)}{(y-z)^2 (y'-z)^2}$ is precisely the "real" piece of the BK/JIMWLK dipole kernel, whose large- z log cancels against the $\frac{1}{(y-z)^2}$ and $\frac{1}{(y'-z)^2}$ pieces — the combination $\frac{(y-y')^2}{(y-z)^2 (y'-z)^2}$ is large- z convergent. Those come from the rows of Group II in which $U(z)$ survives the colour reduction. Here $U(z)$ was eliminated, so this row sees the divergence bare.
- **The cross term** $\frac{1}{\epsilon_{\text{IR}}} \log \frac{k^+}{\Lambda}$ is therefore expected, and is resolved by the same cancellation — it is the statement that the JIMWLK kernel itself must be assembled before the rapidity log is absorbed.

6. Small things still to fix in your expression

1. $\int_{w,w'}$ should be gone — w, w' were eaten by the two delta functions, and z by $\mathcal{G}^{mm'}$. What is left is $\int_{x,x',y,y'}$.
2. $\int_{x,y,y'}$ is missing x' .
3. $U^{ce}(x)$ should be $U^{ec}(x)$ — same index order as its partner $U^{ec}(y)$, otherwise the $[U(y) - U(x)]$ dipole does not form. (Likewise check $U^{e'c}(x')$.)
4. $\epsilon \rightarrow \epsilon_{\text{IR}}$ in $\mathcal{L}$, per §5.

The prefactor $\frac{N_c}{k^+}$ and the phase $e^{-ik(y'-y)} e^{-ik(y'-y)p^+/k^+}$ are now both correct.

7. Summary of the answer to your question

endpoint	singular?	plus prescription?
$p^+ \rightarrow 0$	yes, $\frac{1}{p^+}$ from $\frac{1}{p^+} \delta_{mm'} \delta_{kk'}$	yes — companion is $e^{-i\kappa p^+/k^+}$ with $F(0) = 1$ finite; gives $\log \frac{k^+}{\Lambda}$ plus a finite $[\]_+$ remainder
$p^+ \rightarrow \vee -$ k^+	no pole, but linear growth from $\frac{p^+}{k^{+2}}$	not applicable; handled by the phase, which makes it oscillate and drop

Compare the running-coupling row, where the companion of $\frac{1}{\xi}$ was $\log \frac{\mu^2}{\xi^2 k^2}$ — divergent at the endpoint — and the naive $[\frac{1}{\xi}]_+$ failed, leaving a $\log^2$. Here the companion is bounded, so one prescription suffices and there is no double log.

Verified in `src/rowG2_check.py` (tensor algebra, w, w' Jacobians, $d_\perp$) and `src/rowG2_plus.py` (the infrared nature of the pole, the plus-prescription split, and J_1, J_2 against quadrature).